

Pearson Edexcel

**Level 3 GCE — Predicted Paper — Advanced
Economics A (9EC0/03)**

Paper 3: Microeconomics and Macroeconomics

Synoptic paper. Two case-study sections combining micro and macro.

MARK SCHEME

Total Marks: 100 • Time: 2 hours

(For revision and practice use only.)

General marking guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme — not according to their perception of where the grade boundaries may lie.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved.
- Where some judgement is required, mark schemes will provide the principles by which marks will be awarded and exemplification may be limited.
- When examiners are in doubt regarding the application of the mark scheme to a candidate's response, the team leader must be consulted.
- Crossed-out work should be marked UNLESS the candidate has replaced it with an alternative response.

Advanced-tier expectations

This paper is intended for the highest-attaining A Level candidates and rewards fluent use of theoretical frameworks beyond the core specification. Level 5 responses should demonstrate: (i) explicit use of at least one advanced framework signalled by the question (for example, two-sided markets, direct and indirect network effects, the Lerner condition, dynamic vs static efficiency, Mundell–Fleming, Heckscher–Ohlin, Krugman monopolistic-competition trade, the infant-industry argument, strategic trade policy, general equilibrium effects of tariffs, global value chains); (ii) careful handling of assumptions and of the distinction between partial and general equilibrium; (iii) explicit acknowledgement of the conditions under which the analytical result changes sign; (iv) synoptic integration of micro and macro arguments.

Assessment Objectives — weighting in Paper 3

- AO1: Demonstrate knowledge of terms/concepts and theories.
- AO2: Apply knowledge to problems and issues in familiar and unfamiliar contexts.
- AO3: Analyse issues within economics, showing an understanding of their impact on economic agents.
- AO4: Evaluate economic arguments and use qualitative and quantitative evidence to support informed judgements relating to economic issues.

Levels-based marking for 25-mark questions

Indicative content is provided as a guide and is not exhaustive. Examiners should use the descriptors below to award an overall level, then use a best-fit approach to award a mark within the level.

Level	Marks	Descriptor
1	1–5	Isolated knowledge, largely descriptive. Limited use of extract/figure. Very limited or no analysis; no evaluation.
2	6–10	Sound knowledge of at least two relevant concepts. Some application. Analysis is developed but does not engage with the advanced theoretical frameworks signalled in the question. Evaluation is undeveloped.
3	11–15	Good, accurate knowledge across several concepts. Consistent application. Analysis engages at least briefly with relevant advanced frameworks (e.g. two-sided markets, Mundell–Fleming, strategic trade theory, Lerner condition, new trade theory). Evaluation is attempted.

Level	Marks	Descriptor
4	16–20	Strong, synoptic knowledge. Precise, data-driven application. Analysis demonstrates fluent use of advanced frameworks and shows awareness of assumption sensitivity. Explicit, supported evaluation.
5	21–25	Excellent, integrated, synoptic knowledge. Sophisticated application embedded throughout. Analysis uses advanced frameworks rigorously, distinguishes static from dynamic efficiency and partial from general equilibrium considerations, and handles trade-offs and conditioning assumptions explicitly. Evaluation is prioritised, contextual and leads to a reasoned, supported judgement.

Question 1(a) — Calculation and comment (5 marks)

Ratio of highest-HHI market to lowest-HHI market in Figure 1.

Marks breakdown

AO1	AO2	AO4
1	2	2

Expected working and answer

- Highest HHI: General search services = 8,200. Lowest HHI: Online retail (GMV) = 1,850.
- Ratio: $8,200 / 1,850 = 4.43$ (accept 4.4 or 4.43).
- Comment: search services are around four-and-a-half times more concentrated by HHI than online retail, and are in the “highly concentrated” range ($HHI > 2,500$), suggesting materially different scope for competition policy intervention.

Marking notes

- 1 mark: correct identification of the two HHI values.
- 2 marks: correct method (highest/lowest).
- 2 marks: correct numerical answer with comment linking to concentration thresholds.
- OFR applies if method is correct but a minor slip occurs. No comment caps at 4/5.

Question 1(b) — Discuss (12 marks)

How network effects and economies of scale in data create and sustain market power in digital platforms. Cost and revenue diagram required.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
3	3	3	3

Indicative content

- Direct network effects: utility rises with total users (e.g. messaging, social networks) — demand curve shifts outward as scale grows.
- Indirect network effects in two-sided markets: users on one side (e.g. app developers) value users on the other (consumers) — creates cross-side positive externalities.
- Economies of scale in data: marginal cost of serving one more user approaches zero; algorithms improve with data volume (learning-by-doing).
- Combination produces long-run AC declining over relevant output; natural monopoly tendencies in the core service.
- Lerner condition $(P - MC)/P = 1/|PED|$ — higher perceived differentiation and lower PED imply higher margins.
- Consequences: persistent high margins, weak price-based competition, possible tipping to “winner-takes-most”.

Diagram requirements

- Cost and revenue diagram showing a monopoly or dominant-firm equilibrium: MC, AC declining (often below AR at the profit-maximising Q), AR (= demand), MR. Show price P^* above MC with supernormal profit.
- Alternatively, a two-sided-market diagram showing cross-side externalities is acceptable if correctly labelled.

Evaluation points expected

- Zero-price services — traditional consumer surplus measures are incomplete; welfare trade-off includes data/attention costs.
- Dynamic efficiency: scale can fund R&D and product innovation, raising dynamic consumer welfare even with high static market power.
- Contestability: potential entry by adjacent platforms, disruption through new form factors (e.g. AI assistants).
- Multi-homing: if users can use several services, market power is weaker.
- Differential effects across markets — search has very high tipping (Fig 1, HHI 8,200) vs online retail where multi-homing is easier (HHI 1,850).

Marking notes

- Cap at 9/12 without a properly labelled cost and revenue diagram.
- Top-band responses should explicitly reference the source of persistent market power (network effects + data scale) rather than just describe monopoly outcomes.

Question 1(c) — Examine (8 marks)

Advantages of ex-ante conduct regulation of large digital platforms over ex-post antitrust enforcement.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)
2	2	4

Indicative content

- Speed: ex-post cases typically take 5–10 years by which time dominance is entrenched and rivals may have exited; consumer harm is irreversible.
- Specificity: SMS designation and bespoke conduct requirements can be tailored to the specific sources of harm, rather than shoehorned into generic antitrust categories.
- Deterrence: ex-ante rules align firm conduct before harm occurs; reduces regulatory uncertainty for both platforms and complementors.
- Rivalry preservation: interoperability, data-access and self-preferencing remedies protect a competitive fringe, supporting dynamic efficiency.
- Addresses the under-enforcement problem created by the zero-price “consumer harm” threshold in traditional antitrust.
- Lower evidentiary burden for the regulator vs winning an antitrust case on dominance and abuse.

Application expected

- Extract references to DMCC Act 2024, SMS designation, DMU ex-ante powers.
- Figure data — markets in high-HHI range (search, mobile OS) particularly suited to ex-ante regulation; low-HHI online retail less compelling.

Marking notes

- “Examine” — a strong response will still acknowledge limits of ex-ante (type II errors, regulatory capture, compliance costs on challengers) to reach top of Level 3.
- One-sided advocacy caps at upper Level 2.

Question 1(d) — Evaluate (25 marks)

Factors a regulator should consider when designing pro-competition interventions for firms with Strategic Market Status.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
4	6	7	8

Indicative content — microeconomic factors

- Source of market power: network effects, scale, data advantages, bundling — remedy must map to the mechanism (e.g. interoperability for network effects, data portability for data scale).
- Type I vs Type II errors: cost of over-intervention (discouraging efficient scale) vs under-intervention (entrenched dominance).
- Static vs dynamic efficiency trade-off: conduct rules that reduce static prices may reduce R&D incentives; ensure remedies preserve innovation rents.
- Complementors and multi-sided effects: conduct requirements in one market (e.g. app stores) propagate to many downstream markets.
- Consumer-welfare standard: zero-price markets require a richer welfare framework including data protection and quality.
- Entry conditions and contestability: ensure remedies support the competitive fringe rather than just rearranging market share among incumbents.

Indicative content — macroeconomic factors

- Aggregate productivity and investment: platforms are significant sources of R&D and capex; over-regulation risks weakening a productivity bright spot.
- Regulatory coherence with jurisdictions — EU DMA, US antitrust reform — regulatory arbitrage risk.
- Open-economy effects: UK is a net importer of digital services; incidence of margin reduction largely falls on foreign-owned firms — political economy asymmetries.
- Inflation and consumer prices: digital services deflate CPI contribution; structural changes could have macro-price consequences.
- Labour market: digital jobs are concentrated in London/SE; regulation affecting business location has regional implications.
- Tax revenue: monopoly rents yield significant corporate tax; interaction with OECD Pillar 2.

Evaluation points expected

- Evidence base and burden of proof: ex-ante conduct rules require robust theories of harm; empirical uncertainty argues for reversibility.
- Reversibility: conduct requirements should sunset or be reviewable as markets evolve.
- Regulatory capture and rent-seeking: incumbents may lobby for rules that disadvantage disruptive entrants.
- International coordination — small markets (UK) may find remedies diluted unless aligned with EU/US.
- Interaction with mergers, acquisitions and data policy.
- Judgement: targeted, reviewable interventions anchored in clearly articulated theories of harm are more robust than sweeping structural rules.

Levels apply (see front of mark scheme)

Level	Marks	Descriptor
1	1–5	Isolated knowledge, largely descriptive. Limited use of extract/figure. Very limited or no analysis; no evaluation.

Level	Marks	Descriptor
2	6–10	Sound knowledge of at least two relevant concepts. Some application. Analysis is developed but does not engage with the advanced theoretical frameworks signalled in the question. Evaluation is undeveloped.
3	11–15	Good, accurate knowledge across several concepts. Consistent application. Analysis engages at least briefly with relevant advanced frameworks (e.g. two-sided markets, Mundell–Fleming, strategic trade theory, Lerner condition, new trade theory). Evaluation is attempted.
4	16–20	Strong, synoptic knowledge. Precise, data-driven application. Analysis demonstrates fluent use of advanced frameworks and shows awareness of assumption sensitivity. Explicit, supported evaluation.
5	21–25	Excellent, integrated, synoptic knowledge. Sophisticated application embedded throughout. Analysis uses advanced frameworks rigorously, distinguishes static from dynamic efficiency and partial from general equilibrium considerations, and handles trade-offs and conditioning assumptions explicitly. Evaluation is prioritised, contextual and leads to a reasoned, supported judgement.

Question 1(e) — Evaluate (25 marks)

Micro and macro effects of structural separation (“break-up”) of a dominant digital platform.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
4	6	7	8

Indicative content — microeconomic effects

- Reduces unilateral market power in the separated activity; price or quality improvement in the short run (AR/MR diagram — loss of monopoly rent).
- Eliminates conflicts of interest where the platform competes with its own complementors (e.g. store operator vs app developer).
- Consumer welfare gain in the affected market through greater rivalry and reduced self-preferencing.
- Loss of economies of scope: separated firms may duplicate infrastructure, reducing productive efficiency.
- Loss of dynamic efficiency: vertical integration can accelerate innovation (tight coupling of hardware and software); separation may slow this.
- Impact on complementors: app developers, advertisers, small sellers — heterogeneous; some gain from greater parity, some lose from reduced platform-wide investment.
- Labour market: fragmentation may create jobs in duplicated functions but reduce high-skill engineering scale.

Indicative content — macroeconomic effects

- Aggregate productivity: unclear sign. Short-run disruption vs long-run gains from competition; evidence from 20th-century break-ups (AT&T) ambiguous.
- Investment: break-up reduces scale-based investment; offset by increased rival investment if markets become more contestable.
- Consumer prices and CPI: digital services prices may fall slightly; quality effects harder to measure.
- International effects: UK break-up of a global firm has limited global effect given UK’s share of user base; likely effective only if coordinated with EU/US.
- Exchange rate/BoP: digital services are significant UK exports; disruption could affect services trade balance.
- Tax revenues: possible short-run reduction as separated entities report lower combined profit; longer-run gains if new entrants emerge.

Diagram suggestions

- Monopoly → competitive equilibrium diagram: movement from MC=MR with monopoly profit to more competitive outcome.
- Production possibility / efficiency frontier showing trade-off between static and dynamic efficiency.

Evaluation points expected

- Remedy design is critical: horizontal vs vertical separation; behavioural vs structural; one-off vs phased.
- Irreversibility: structural remedies are hard to reverse; should be a last resort after conduct remedies fail.
- International scope: unilateral UK separation is unlikely to be effective given global scale.
- Incentive effects: may deter future investment by successful firms (chilling effect) — or spur it by clarifying the competitive “rules of the game”.

- Heterogeneity across platforms: break-up is more defensible in markets with weak economies of scope and strong vertical leverage.
- Relative merits of alternative remedies: interoperability mandates, data access, pro-competition choice architecture.
- Supported judgement: structural separation has a narrow range of circumstances where net welfare is positive; for most cases, well-designed ex-ante conduct remedies dominate.

Levels apply (see front of mark scheme)

Question 2(a) — Explain (5 marks)

Direction of change in global trade openness since 2018 from Figure 2.

Marks breakdown

AO1 (K)	AO2 (App)
2	3

Indicative content

- Average US tariff on China imports rose from 3.1% (2018) to 17.2% (2025) — roughly a 5.5× increase, indicating a substantial increase in cross-bloc trade frictions.
- Share of FDI flowing to geopolitically aligned partners rose from 65% to 86% — FDI is being redirected along geopolitical lines ('friend-shoring').
- Cumulative G20 industrial-policy announcements rose from 380 (2018) to 1,920 (2025) — fivefold increase; indicates a systemic shift from market-led to state-led industrial strategy.
- Interpretation: taken together, the data suggest a consistent and accelerating shift toward geoeconomic fragmentation — a reduction in the openness of the rules-based multilateral trading system.

Marking notes

- 1–2 marks: simple description of one indicator.
- 3–4 marks: two indicators correctly interpreted together.
- 5 marks: all three indicators plus overarching interpretation identifying fragmentation/friend-shoring pattern.

Question 2(b) — Examine (8 marks)

Effects on profitability of a UK exporter of intermediates of a 15% tariff on exports to a major market, with cost/revenue diagram.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)
2	2	4

Indicative content

- Incidence of tariff depends on relative elasticities of export supply and import demand — burden shared between UK exporter and foreign importer.
- If importer market is competitive and PED is high, exporter bears more of the tariff — effective price received falls, reducing revenue per unit.
- If UK firm has product differentiation or switching costs, can pass through more of the tariff — preserves margin but loses volume.
- Fixed costs and scale economies: lower volume raises AC — margin pressure compounds.
- Adjustment options: relocation (incurring sunk investment costs), switching to alternative markets (if comparable), vertical integration upstream.
- Global value chain dimension: if the firm imports inputs, reciprocal tariffs compound the cost shock.

Diagram requirements

- Cost and revenue diagram showing a firm with AR, MR, AC, MC. A tariff shifts AR downward (the demand curve facing the UK exporter drops) or raises AC depending on how

it is modelled. Show original supernormal or normal profit contracting, with lower Q^* and/or lower P^* .

- Alternatively (acceptable): a tariff-incidence diagram in the export market showing the wedge between price paid by importer and price received by exporter.

Marking notes

- Cap at 5/8 without an appropriate diagram.
- Reward candidates who distinguish incidence from nominal tariff rate, and who engage with PED and scale effects.

Question 2(c) — Discuss (12 marks)

Micro and macro effects on a small open economy (UK) of sustained fragmentation of global value chains.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
3	3	3	3

Microeconomic effects

- Rising input costs: fragmented GVCs raise transaction and logistics costs; exporters face higher AC.
- Lost specialisation gains: reduced ability to exploit comparative advantage (Heckscher–Ohlin losses); reduced scale in intermediate-good production (Krugman monopolistic-competition losses).
- Firm heterogeneity: larger firms can adjust supply chains; SMEs bear disproportionate costs — distributional effect within the tradable sector.
- Specific sectors hit hardest: automotive, pharmaceuticals, electronics, chemicals — all deeply integrated into multi-country supply chains.
- Consumer welfare: reduced variety of goods (Krugman ‘love-of-variety’), higher prices.

Macroeconomic effects

- LRAS: negative shift as productive efficiency declines — permanent output loss.
- Inflation: supply-side cost push — structurally higher inflation for a given AD path, complicating monetary policy.
- Balance of payments: UK is services-surplus, goods-deficit — composition effects; services trade may also be fragmented (e.g. data-localisation rules).
- Exchange rate: weaker tradable-sector competitiveness could depreciate sterling; but also loss of safe-asset status in a fragmented world could reprice gilts (link to debt-sustainability).
- Employment: displacement in exposed manufacturing/services; reallocation frictions.

Evaluation points expected

- Offsetting factors: re-shoring can raise domestic value added and employment in strategic sectors.
- Magnitude depends on whether the UK can pivot toward a specific bloc (e.g. deeper EU or US integration) or remains between blocs.
- Time horizon: short-run adjustment costs vs long-run reallocation gains.
- Small-economy vulnerability: unlike the US or EU, the UK cannot build globally competitive scale in a single bloc for all strategic sectors.
- Services-heavy economy: UK comparative advantage in tradable services may be relatively insulated — depends on whether services face parallel fragmentation.

- Overall: fragmentation likely net negative for a small, open, services-exporting economy, but net position depends heavily on policy response.

Question 2(d) — Evaluate (25 marks)

Micro and macro consequences for an advanced economy of using large-scale strategic industrial policy to respond to geoeconomic fragmentation.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
4	6	7	8

Indicative content — microeconomic effects

- Subsidies and local-content requirements correct for positive externalities in strategic sectors (learning spillovers, knowledge production).
- Infant-industry argument: temporary protection can enable firms to achieve scale, if learning curves are steep and sunset clauses are credible.
- Market failure analysis: environmental externalities (climate), security externalities (critical supply chains), agglomeration externalities.
- Distortion costs: subsidies create deadweight losses (Harberger triangles) in supported sectors; tax-financing creates additional DWL.
- Rent-seeking and capture risk: information asymmetries between regulator and firm favour incumbents.
- Cluster/agglomeration effects: successful industrial policy can generate persistent productive ecosystems; failed policies sink political and fiscal capital.

Indicative content — macroeconomic effects

- Aggregate demand: subsidy-funded investment raises AD in the short run; multiplier effects largest where spare capacity exists.
- LRAS: positive if subsidy raises productive capacity in genuinely high-externality sectors; negative if resources are diverted from higher-productivity activities (resource misallocation).
- Fiscal position: significant upfront cost; crowding out risk if financed by borrowing; link to gilt yields and debt-sustainability.
- Trade balance and exchange rate: local-content requirements shift domestic demand toward domestic production — narrows trade deficit but may appreciate currency.
- Retaliation: subsidy races risk zero-sum outcomes (US IRA → EU Chips Act → UK response). Game-theoretic analysis: non-cooperative Nash equilibrium dominates Pareto-optimal multilateralism.
- Inflation: short-run sector-specific wage pressures in targeted sectors; demand effects of subsidy stimulus; crowding in of skilled labour.

Theoretical frameworks expected

- Strategic trade theory (Brander–Spencer): subsidies can shift rents in imperfectly competitive global industries under specific conditions.
- Infant industry (Hamilton–List): temporary protection to achieve scale.
- Public economics: Pigouvian subsidies for externality correction vs deadweight-loss-maximising universal subsidies.
- General equilibrium effects: resources drawn to subsidised sector raise wages/input costs in unsubsidised sectors.

Evaluation points expected

- Targeting precision: only defensible for sectors with genuine, measurable externalities — weak case for general industrial policy.
- Sunset mechanisms and evaluation — credibility of time-limits is often weak.

- International coordination vs retaliation: collective bargaining mechanisms (OECD, WTO) may reduce subsidy-race waste.
- Fiscal space: feasibility depends on debt sustainability (link to r vs g).
- Political economy: beneficial sectors are visible; deadweight costs are diffuse — capture risk is inherent.
- Counterfactual — absent industrial policy, does fragmentation still damage the economy? Strategic policy may be a second-best response to a bad first-best.
- Judgement: targeted, time-limited, externality-based industrial policy with international coordination can be welfare-improving; undisciplined subsidy races are likely net welfare-negative for the global economy and often for the participating economy.

Levels apply (see front of mark scheme)

Question 2(e) — Evaluate (25 marks)

Case for UK unilateral trade liberalisation and multilateral re-engagement as a response to global fragmentation.

Marks breakdown

AO1 (K)	AO2 (App)	AO3 (An)	AO4 (Ev)
4	6	7	8

Indicative content — case for unilateral liberalisation and multilateralism

- Ricardian comparative advantage: unilateral reduction in import barriers improves UK welfare even if trading partners do not reciprocate (the “terms-of-trade” caveat aside for a small open economy).
- Input-cost reduction: lower tariffs on imported intermediates raise the competitiveness of UK exporters.
- Consumer surplus: lower tariffs reduce prices and expand variety — welfare gains identifiable in general equilibrium models.
- Dynamic efficiency: import competition disciplines domestic market power — raises productivity in the long run.
- Small-economy specialisation gains are large relative to GDP — the UK has more to lose from fragmentation than the US or China.
- Multilateral institutions (WTO, plurilateral deals) lock in reciprocal openness and reduce the risk of future escalation — they are public goods under-produced by any single state.
- Credible commitment device: independent trade policy frameworks reduce incentives for rent-seeking by domestic producers.

Indicative content — case against pure unilateralism

- Adjustment costs: import liberalisation displaces workers in previously protected sectors; labour-market frictions can make transitional losses significant and politically costly (e.g. “China shock” literature).
- Strategic sectors: complete openness may increase dependence on unreliable suppliers (critical minerals, semiconductors, pharmaceuticals).
- Reciprocity argument: unilateral openness offers domestic concessions without extracting foreign openness; reduces bargaining power in future negotiations.
- National security and resilience arguments for some strategic protection.
- Climate policy: border adjustments (CBAM) may be needed to prevent carbon leakage — clashes with pure free-trade stance.
- Fragmented world: multilateral re-engagement requires willing partners — the WTO dispute-settlement mechanism is partially dysfunctional; limits the practical return to openness.

Application expected

- Use of Figure 2: tariffs rising, FDI redirecting, industrial-policy announcements proliferating — the international context for any UK strategy.
- Use of extract: “fragmentation of global value chains imposes permanent efficiency losses”; UK is “outside the major trading blocs”.
- Small-open-economy framework: UK has limited ability to shift terms of trade; gains from openness are proportionally large.

Evaluation points expected

- Pure unilateral liberalisation vs pragmatic approach: target strategic sectors for modest resilience investment while liberalising broadly.

- Sequencing: liberalise first where adjustment costs are lowest; protect retraining and regional adjustment.
- Multilateral re-engagement is necessary but not sufficient — requires willing partners and credible commitments.
- Second-best considerations: if other economies subsidise heavily, UK unilateralism may be welfare-negative in specific sectors.
- Time-inconsistency: unilateral commitments need credible institutions (independent trade body, legislated tariff schedules).
- Distributional consequences — need a coherent domestic adjustment strategy, otherwise political sustainability is fragile.
- Supported judgement: for a small, services-heavy, open economy, the balance of evidence favours broad-based liberalisation coupled with targeted resilience policies in genuinely strategic sectors and active multilateral leadership — pure unilateral openness without complementary policies is politically and economically risky.

Levels apply (see front of mark scheme)

TOTAL MARKS AVAILABLE: 100

END OF MARK SCHEME